Equations

CHEMISTRY REFERENCE

Measurement

Density	$D = \frac{m}{V}$
Percent error	$\%\ \text{error} = \frac{\lvert \text{experimental} - \text{accepted} \rvert}{\text{accepted}} \times 100$
Percent composition	$\%\ \text{element} = \frac{\text{mass of element}}{\text{mass of compound}} \times 100$
Average atomic mass	$(\text{mass}_1 \times \text{abundance}_1) + (\text{mass}_2 \times \text{abundance}_2) + \ldots$ abundance as a decimal

The Mole

Moles from mass	$n = \frac{m}{M}$ n moles · m mass (g) · M molar mass (g/mol)
Particles	$\text{particles} = n \times N_A$

Gas Laws T in kelvin

Boyle's law	$P_1V_1 = P_2V_2$
Charles's law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay-Lussac's law	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Combined gas law	$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
Avogadro's law	$\frac{V_1}{n_1} = \frac{V_2}{n_2}$
Ideal gas law	$PV = nRT$
Dalton's law	$P_{\text{total}} = P_1 + P_2 + P_3 + \ldots$

Solutions

Molarity	$M = \frac{\text{moles of solute}}{\text{liters of solution}}$
Dilution	$M_1V_1 = M_2V_2$

Thermochemistry

Heat	$q = mc\Delta T$
Temperature change	$\Delta T = T_{\text{final}} - T_{\text{initial}}$

Acids & Bases

pH	$\text{pH} = -\log[H^+] \quad [H^+] = 10^{-\text{pH}}$
pOH	$\text{pOH} = -\log[OH^-] \quad [OH^-] = 10^{-\text{pOH}}$
pH and pOH	$\text{pH} + \text{pOH} = 14$ at 25 °C
Titration (1 : 1)	$M_aV_a = M_bV_b$

Light

Wave speed	$c = \lambda\nu$
Photon energy	$E = h\nu$

Nuclear

Half-life	$\text{amount left} = \text{initial amount} \times (\frac{1}{2})^n$
	$n = \frac{\text{time elapsed}}{\text{half-life}}$

Constants & Conversions

CHEMISTRY REFERENCE

Constants

Avogadro's number	$N_A = 6.022 \times 10^{23}$ particles/mol
Gas constant	R = 0.0821 L·atm/(mol·K)
	R = 8.314 L·kPa/(mol·K)
	R = 62.4 L·mmHg/(mol·K)
Speed of light	$c = 3.00 \times 10^{8}$ m/s
Planck's constant	$h = 6.626 \times 10^{-34}$ J·s
Specific heat of water	c = 4.184 J/(g·°C)
STP	0 °C (273 K) and 1 atm
Molar volume at STP	22.4 L/mol

Metric Prefixes

Prefix	Symbol	Factor
giga	**G**	10^{9}
mega	**M**	10^{6}
kilo	**k**	10^{3}
— base unit —		1
deci	**d**	10^{-1}
centi	**c**	10^{-2}
milli	**m**	10^{-3}
micro	**μ**	10^{-6}
nano	**n**	10^{-9}

Conversions

Pressure	1 atm = 760 mmHg = 760 torr = 101.3 kPa
Temperature	K = °C + 273
Volume	1 mL = 1 cm^3 1 L = 1000 mL
Length	1 in = 2.54 cm
Mass	1 lb = 453.6 g 1 kg = 1000 g
Energy	1 cal = 4.184 J 1 kJ = 1000 J

Dimensional Analysis

$$\boxed{\text{given}\\ \text{number + unit}} \times \frac{\boxed{\text{new unit}}}{\boxed{\text{unit to cancel}}} \times \frac{\boxed{\text{new unit}}}{\boxed{\text{unit to cancel}}} = \boxed{\text{answer}\\ \text{number + unit}}$$

1. Write what you are given, with its unit.
2. Multiply by a conversion factor written so the unit you have is on the bottom.
3. Cancel units. Repeat until only the unit you want is left.
4. Multiply across the top, divide by everything on the bottom.

Example $1.75 \text{ mol } H_2O \times \frac{18.02 \text{ g } H_2O}{1 \text{ mol } H_2O} = 31.5 \text{ g } H_2O$